

Revisiting SIF rules abstraction with SPARQL for querying BioPAX - Supplementary Material: BioPAX to SIF SPARQL Queries

1 controls-state-change-of

Level 1: SIF rule description from PathwayCommons

First protein controls a reaction that changes the state of the second protein.

Level 2: Paxtools pattern controlsStateChange() description

Pattern for a EntityReference has a member PhysicalEntity that is controlling a state change reaction of another EntityReference.

```
CONSTRUCT {  
  ?enzymeRef abstraction:ControlsStateChangeOf ?protRef  
}  
WHERE {  
  ?reaction rdf:type/(rdfs:subClassOf*) bp3:Conversion .  
  
  ?catalysis bp3:controlled ?reaction .  
  ?catalysis (rdf:type/rdfs:subClassOf*) bp3:Control .  
  ?catalysis bp3:controller ?enzyme .  
  ?enzyme rdf:type bp3:Protein .  
  ?enzyme bp3:entityReference ?enzymeRef .  
  ?enzymeRef rdf:type/(rdfs:subClassOf*) bp3:EntityReference .  
  
  ?reaction bp3:left ?protLeft .  
  ?protLeft rdf:type bp3:Protein .  
  ?protLeft bp3:entityReference ?protRef .  
  
  ?reaction bp3:right ?protRight .  
  ?protRight rdf:type bp3:Protein .  
  ?protRight bp3:entityReference ?protRef .  
  
  ?protRef rdf:type/(rdfs:subClassOf*) bp3:EntityReference .  
  
  ?protRight bp3:feature ?feature .  
  ?feature (rdf:type/rdfs:subClassOf*) bp3:EntityFeature .  
  FILTER (?enzymeRef != ?protRef)  
}
```

2 controls-transport-of

Level 1: SIF rule description from PathwayCommons

First protein controls a reaction that changes the cellular location of the second protein.

Level 2: Paxtools pattern controlsTransport() description

Pattern for a ProteinReference has a member PhysicalEntity that is controlling a transportation of another Protein-Reference.

```
CONSTRUCT {  
  ?enzymeRef abstraction:ControlsTransportOf ?protRef  
}  
WHERE {  
  ?reaction rdf:type/(rdfs:subClassOf*) bp3:Conversion .  
  
  ?reaction bp3:left ?protLeft .  
  ?protLeft rdf:type bp3:Protein .  
  ?protLeft bp3:entityReference ?protRef .  
  
  ?reaction bp3:right ?protRight .  
  ?protRight rdf:type bp3:Protein .  
  ?protRight bp3:entityReference ?protRef .  
  
  ?protRef rdf:type bp3:ProteinReference .  
  
  # Define cellular locations for Protein 1 and Protein 2  
  ?protLeft bp3:cellularLocation ?cellularLocVocab1 .  
  ?protRight bp3:cellularLocation ?cellularLocVocab2 .  
  
  ?catalysis bp3:controlled ?reaction .  
  ?catalysis (rdf:type/rdfs:subClassOf*) bp3:Control .  
  ?catalysis bp3:controller ?enzyme .  
  ?enzyme rdf:type bp3:Protein .  
  ?enzyme bp3:entityReference ?enzymeRef .  
  ?enzymeRef rdf:type bp3:ProteinReference .  
  
  # Ensure cellular locations of protein 1 and protein 2 are different  
  FILTER (?cellularLocVocab1 != ?cellularLocVocab2)  
  # Ensure the enzyme is not the transported protein  
  FILTER (?enzymeRef != ?protRef)  
}
```

3 controls-phosphorylation-of

Level 1: SIF rule description from PathwayCommons

First protein controls a reaction that changes the phosphorylation status of the second protein.

Level 2: Paxtools pattern controlsPhosphorylation() description

No function description in Paxtools

Explicated Level 2

Pattern for an EntityReference that has a memberEntity that controls a state change reaction that phosphorylates the memberEntity of another EntityReference.

```
CONSTRUCT {  
  ?enzymeRef abstraction:ControlsPhosphorylationOf ?protRef  
}  
WHERE {  
  ?reaction rdf:type/(rdfs:subClassOf*) bp3:Conversion .  
  
  ?catalysis bp3:controlled ?reaction .  
  ?catalysis rdf:type/(rdfs:subClassOf*) bp3:Control .  
  ?catalysis bp3:controller ?enzyme .  
  ?enzyme rdf:type bp3:Protein .  
  ?enzyme bp3:entityReference ?enzymeRef .  
  ?enzymeRef rdf:type/(rdfs:subClassOf*) bp3:EntityReference .  
  
  ?reaction bp3:left ?protLeft .  
  ?protLeft rdf:type bp3:Protein .  
  ?protLeft bp3:entityReference ?protRef .  
  
  ?reaction bp3:right ?protRight .  
  ?protRight rdf:type bp3:Protein .  
  ?protRight bp3:entityReference ?protRef .  
  
  ?protRef rdf:type/(rdfs:subClassOf*) bp3:EntityReference .  
  
  # Specify that ?protRight is phosphorylated  
  ?protRight bp3:feature ?feature .  
  ?feature rdf:type bp3:ModificationFeature .  
  ?feature bp3:modificationType ?modificationType .  
  ?modificationType rdf:type bp3:SequenceModificationVocabulary .  
  ?modificationType bp3:term ?modificationTerm .  
  FILTER(CONTAINS(LCASE(?modificationTerm), "phospho"))  
}
```

4 controls-expression-of

Level 1: SIF rule description from PathwayCommons

First protein controls a conversion or a template reaction that changes expression of the second protein.

Level 2: Paxtools pattern controlsExpressionWithTemplateReac() description

Finds transcription factors that trans-activate or trans-inhibit an entity.

Level 2: Paxtools pattern controlsExpressionWithConversion() description

Finds the cases where transcription relation is shown using a Conversion instead of a TemplateReaction.

Explicated Level 2

Pattern for an EntityReference that has a memberEntity that controls a TemplateReaction or a Conversion reaction that changes the expression of the memberEntity of another EntityReference.

```

CONSTRUCT {
  ?enzymeRef abstraction:ControlsExpressionOf ?productRef
}
WHERE {
# Case where the reaction is a Conversion
{
  ?reaction rdf:type/(rdfs:subClassOf*) bp3:Conversion .
  ?reaction bp3:right ?product .
  ?product rdf:type bp3:Protein .
  ?reaction bp3:displayName ?reactionName .
  ?catalysis bp3:controlled ?reaction .
  ?catalysis rdf:type/(rdfs:subClassOf*) bp3:Control .
  ?product bp3:entityReference ?productRef .
  ?productRef rdf:type bp3:ProteinReference .
  ?catalysis bp3:controller ?enzyme .
  ?enzyme rdf:type bp3:Protein .
  ?enzyme bp3:entityReference ?enzymeRef .
  ?enzymeRef rdf:type bp3:ProteinReference .
  FILTER(CONTAINS(LCASE(?reactionName), "expression"))
}

UNION

# Case where the reaction is a TemplateReaction
{
  ?reaction rdf:type/(rdfs:subClassOf*) bp3:TemplateReaction .
  ?reaction bp3:product ?product .
  ?product rdf:type bp3:Protein .
  ?catalysis bp3:controlled ?reaction .
  ?catalysis rdf:type bp3:TemplateReactionRegulation .
  ?product bp3:entityReference ?productRef .
  ?productRef rdf:type bp3:ProteinReference .
  ?catalysis bp3:controller ?enzyme .
  ?enzyme rdf:type bp3:Protein .
  ?enzyme bp3:entityReference ?enzymeRef .
  ?enzymeRef rdf:type bp3:ProteinReference .
}
}

```

5 catalysis-precedes

Level 1: SIF rule description from PathwayCommons

First protein controls a reaction whose output molecule is input to another reaction controlled by the second protein.

Level 2: Paxtools pattern controlsMetabolicCatalysis() description

Pattern for detecting two EntityReferences are controlling consecutive reactions, where output of one reaction is input to the other.

```
CONSTRUCT {  
  ?enzymeRefA abstraction:CatalysisPrecedes ?enzymeRefB  
}  
WHERE {  
  ?reaction1 rdf:type/(rdfs:subClassOf*) bp3:Conversion .  
  ?catalysis1 bp3:controlled ?reaction1 .  
  ?catalysis1 (rdf:type/rdfs:subClassOf*) bp3:Control .  
  ?catalysis1 bp3:controller ?enzymeA .  
  ?enzymeA rdf:type bp3:Protein .  
  ?enzymeA bp3:entityReference ?enzymeRefA .  
  ?enzymeRefA rdf:type/(rdfs:subClassOf*) bp3:EntityReference .  
  
  ?reaction1 bp3:right ?connectingMolecule .  
  ?reaction2 bp3:left ?connectingMolecule .  
  ?connectingMolecule rdf:type bp3:SmallMolecule .  
  
  ?reaction2 rdf:type/(rdfs:subClassOf*) bp3:Conversion .  
  ?catalysis2 bp3:controlled ?reaction2 .  
  ?catalysis2 (rdf:type/rdfs:subClassOf*) bp3:Control .  
  ?catalysis2 bp3:controller ?enzymeB .  
  ?enzymeB rdf:type bp3:Protein .  
  ?enzymeB bp3:entityReference ?enzymeRefB .  
  ?enzymeRefB rdf:type/(rdfs:subClassOf*) bp3:EntityReference .  
  
  FILTER (?enzymeRefA != ?enzymeRefB)  
}
```

6 in-complex-with

Level 1: SIF rule description from PathwayCommons

Proteins are members of the same complex.

Level 2: Paxtools pattern inComplexWith() description

Two proteins have states that are members of the same complex. Handles nested complexes and homologies. Also guarantees that relationship to the complex is through different direct complex members.

Explicated Level 2

Pattern for two ProteinReferences that have member PhysicalEntities that are members of the same complex.

```

CONSTRUCT {
  ?prot1Ref abstraction:InComplexWith ?prot2Ref
}
WHERE {
  # Case 1: Complex with several different proteins
  {
    ?complex rdf:type/(rdfs:subClassOf*) bp3:Complex .
    ?complex bp3:component ?prot1 .
    ?complex bp3:component ?prot2 .
    ?prot1 rdf:type bp3:Protein .
    ?prot2 rdf:type bp3:Protein .

    # Avoid self-pairing and duplicate pairs
    FILTER (STR(?prot1) < STR(?prot2))
  }

  UNION

  # Case 2: Same protein with stoichiometry > 1
  {
    ?complex rdf:type/(rdfs:subClassOf*) bp3:Complex .
    # Check for stoichiometry > 1
    ?complex bp3:componentStoichiometry ?stoich .
    ?stoich bp3:physicalEntity ?entity1 .
    ?stoich bp3:stoichiometricCoefficient ?coeff .
    FILTER (?coeff > 1)
    ?complex bp3:component ?prot1 .
    ?prot1 rdf:type bp3:Protein .

    # For self-pairing, use the same entity
    BIND (?prot1 AS ?prot2)
  }

  ?prot1 bp3:entityReference ?prot1Ref .
  ?prot2 bp3:entityReference ?prot2Ref .
  ?prot1Ref rdf:type bp3:ProteinReference .
  ?prot2Ref rdf:type bp3:ProteinReference .
}

```


7 interacts-with

Level 1: SIF rule description from PathwayCommons

Proteins are participants of the same MolecularInteraction.

Level 2: Paxtools pattern molecularInteraction() description

Constructs a pattern where first and last molecules are participants of a MolecularInteraction.

Explicated Level 2

Pattern for two EntityReferences that have member PhysicalEntities that are members of the same MolecularInteraction.

```
CONSTRUCT {  
  ?participant1Ref abstraction:InteractsWith ?participant2Ref  
}  
WHERE {  
  VALUES ?participantType { bp3:Protein bp3:SmallMolecule }  
  # Core molecular interaction pattern  
  ?MolecularInteraction rdf:type/(rdfs:subClassOf*) bp3:MolecularInteraction .  
  ?MolecularInteraction bp3:participant ?participant1 .  
  ?participant1 rdf:type ?participantType .  
  ?MolecularInteraction bp3:participant ?participant2 .  
  ?participant2 rdf:type ?participantType .  
  
  ?participant1 bp3:entityReference ?participant1Ref .  
  ?participant1Ref rdf:type/(rdfs:subClassOf*) bp3:EntityReference .  
  ?participant2 bp3:entityReference ?participant2Ref .  
  ?participant2Ref rdf:type/(rdfs:subClassOf*) bp3:EntityReference .  
  
  FILTER (STR(?participant1Ref) < STR(?participant2Ref))  
}
```

8 neighbor-of

Level 1: SIF rule description from PathwayCommons

Proteins are participants or controllers of the same interaction.

Level 2: Paxtools pattern neighborOf() description

Constructs a pattern where first and last proteins are related through an interaction. They can be participants or controllers. No limitation.

Explicated Level 2

Pattern for two EntityReferences that have member PhysicalEntities that are members of the same Interaction. Member PhysicalEntities can be participants or controllers of the interaction.

```
CONSTRUCT {  
  ?prot1Ref abstraction:NeighborOf ?prot2Ref  
}  
WHERE {  
  ?reaction rdf:type/(rdfs:subClassOf*) bp3:Interaction .  
  
  ?reaction ((^bp3:controlled/bp3:controller)|bp3:left|bp3:right) ?prot1 .  
  ?prot1 rdf:type bp3:Protein .  
  ?prot1 bp3:entityReference ?prot1Ref .  
  ?prot1Ref rdf:type bp3:ProteinReference .  
  ?reaction ((^bp3:controlled/bp3:controller)|bp3:left|bp3:right) ?prot2 .  
  ?prot2 rdf:type bp3:Protein .  
  ?prot2 bp3:entityReference ?prot2Ref .  
  ?prot2Ref rdf:type bp3:ProteinReference .  
  FILTER (STR(?prot1Ref) < STR(?prot2Ref))  
}
```

9 consumption-controlled-by

Level 1: SIF rule description from PathwayCommons

The small molecule is consumed by a reaction that is controlled by a protein

Level 2: Paxtools pattern description

No description

Explicated Level 2

Pattern for a SmallMoleculeReference that has a member PhysicalEntity that is a small molecule consumed by a reaction that is controlled by a protein that is linked to a ProteinReference.

```
CONSTRUCT {  
  ?reactantRef abstraction:ConsumptionControlledBy ?enzymeRef  
}  
WHERE {  
  ?reaction rdf:type/(rdfs:subClassOf*) bp3:Conversion .  
  ?reaction bp3:left ?reactant .  
  ?reactant rdf:type bp3:SmallMolecule .  
  ?reactant bp3:entityReference ?reactantRef .  
  ?reactantRef rdf:type bp3:SmallMoleculeReference .  
  ?reaction bp3:right ?product .  
  ?product rdf:type bp3:SmallMolecule .  
  ?catalysis bp3:controlled ?reaction .  
  ?catalysis bp3:controller ?enzyme .  
  ?enzyme rdf:type bp3:Protein .  
  ?enzyme bp3:entityReference ?enzymeRef .  
  ?enzymeRef rdf:type bp3:ProteinReference .  
}
```

10 controls-production-of

Level 1: SIF rule description from PathwayCommons

The protein controls a reaction of which the small molecule is an output.

Level 2: Paxtools pattern description

No description

Explicated Level 2

Pattern for a ProteinReference that has a member PhysicalEntity that controls a reaction that produces a small molecule (a product) linked to a SmallMoleculeReference.

```
CONSTRUCT {  
  ?enzymeRef abstraction:ControlsProductionOf ?productRef  
}  
WHERE {  
  ?reaction rdf:type/(rdfs:subClassOf*) bp3:Conversion .  
  ?reaction bp3:left ?reactant .  
  ?reactant rdf:type bp3:SmallMolecule .  
  ?reaction bp3:right ?product .  
  ?product rdf:type bp3:SmallMolecule .  
  ?product bp3:entityReference ?productRef .  
  ?productRef rdf:type bp3:SmallMoleculeReference .  
  ?catalysis bp3:controlled ?reaction .  
  ?catalysis bp3:controller ?enzyme .  
  ?enzyme rdf:type bp3:Protein .  
  ?enzyme bp3:entityReference ?enzymeRef .  
  ?enzymeRef rdf:type bp3:ProteinReference .  
}
```

11 controls-transport-of-chemical

Level 1: SIF rule description from PathwayCommons

The protein controls a reaction that changes cellular location of the small molecule.

Level 2: Paxtools pattern controlsTransportOfChemical() description

Pattern for a ProteinReference has a member PhysicalEntity that is controlling a reaction that changes cellular location of a small molecule.

Explicated Level 2

Pattern for a ProteinReference that has a member PhysicalEntity that is controlling a reaction that changes the location of a small molecule linked to a SmallMoleculeReference.

```
CONSTRUCT {  
  ?enzymeRef abstraction:ControlsTransportOfChemical ?SmallMoleculeRef  
}  
WHERE {  
  ?reaction rdf:type/(rdfs:subClassOf*) bp3:Conversion .  
  ?reaction bp3:left ?SmallMolecule1 .  
  ?SmallMolecule1 rdf:type bp3:SmallMolecule .  
  ?SmallMolecule1 bp3:cellularLocation ?cellularLocVocab1 .  
  ?SmallMolecule1 bp3:entityReference ?SmallMoleculeRef .  
  
  ?reaction bp3:right ?SmallMolecule2 .  
  ?SmallMolecule2 rdf:type bp3:SmallMolecule .  
  ?SmallMolecule2 bp3:cellularLocation ?cellularLocVocab2 .  
  ?SmallMolecule2 bp3:entityReference ?SmallMoleculeRef .  
  
  ?SmallMoleculeRef rdf:type bp3:SmallMoleculeReference .  
  
  ?catalysis bp3:controlled ?reaction .  
  ?catalysis rdf:type/(rdfs:subClassOf*) bp3:Control .  
  ?catalysis bp3:controller ?enzyme .  
  ?enzyme rdf:type bp3:Protein .  
  ?enzyme bp3:entityReference ?enzymeRef .  
  ?enzymeRef rdf:type bp3:ProteinReference .  
  
  FILTER(?cellularLocVocab1 != ?cellularLocVocab2)  
}
```

12 chemical-affects

Level 1: SIF rule description from PathwayCommons

A small molecule has an effect on the protein state.

Level 2: Paxtools pattern `chemicalAffectsProteinThroughBinding()` description

A small molecule is in a complex with a protein.

Level 2: Paxtools pattern `chemicalAffectsProteinThroughControl()` description

A small molecule controls an interaction of which the protein is a participant.

Explicated Level 2

Pattern for a `SmallMoleculeReference` that has a member `PhysicalEntity` that changes the state of a protein linked to a `ProteinReference`, either by controlling a `Interaction` of which the protein is a participant or by binding with the protein in a complex.

```

CONSTRUCT {
  ?chemicalCatalyzerRef abstraction:ChemicalAffects ?proteinRef .
}
WHERE {
  # the small molecule has an effect on the protein state through binding
  {
    ?complex rdf:type bp3:Complex .
    ?complex bp3:component ?proteinComponent .
    ?proteinComponent rdf:type bp3:Protein .
    # complete modification check
    ?proteinComponent bp3:feature ?feature .
    ?feature rdf:type/(rdfs:subClassOf*) bp3:EntityFeature .
    ?complex bp3:component ?chemicalCatalyzer .
    ?chemicalCatalyzer rdf:type bp3:SmallMolecule .
    ?chemicalCatalyzer bp3:entityReference ?chemicalCatalyzerRef .
    ?chemicalCatalyzerRef rdf:type bp3:SmallMoleculeReference .

    BIND(?proteinComponent AS ?protein1)
  }

  UNION

  # the small molecule has an effect on the protein state by controlling a Conversion reaction
  {
    ?reaction rdf:type/(rdfs:subClassOf*) bp3:Conversion .
    ?reaction bp3:left ?protLeft .
    ?protLeft rdf:type bp3:Protein .
    ?protLeft bp3:entityReference ?protRef .

    ?reaction bp3:right ?protRight .
    ?protRight rdf:type bp3:Protein .
    ?protRight bp3:entityReference ?protRef .

    ?protRef rdf:type bp3:ProteinReference .

    ?catalysis bp3:controlled ?reaction .
    ?catalysis rdf:type/(rdfs:subClassOf*) bp3:Control .
    ?catalysis bp3:controller ?chemicalCatalyzer .
    ?chemicalCatalyzer rdf:type bp3:SmallMolecule .
    ?chemicalCatalyzer bp3:entityReference ?chemicalCatalyzerRef .
    ?chemicalCatalyzerRef rdf:type bp3:SmallMoleculeReference .

    # complete modification check
    ?protRight bp3:feature ?feature .
    ?feature rdf:type/(rdfs:subClassOf*) bp3:EntityFeature .

    BIND(?protLeft AS ?protein2)
  }

  BIND( COALESCE(?protein1, ?protein2) AS ?protein )
  ?protein bp3:entityReference ?proteinRef .
  ?proteinRef rdf:type bp3:ProteinReference .
}

```

13 reacts-with

Level 1: SIF rule description from PathwayCommons

Small molecules are input to a biochemical reaction.

Level 2: Paxtools pattern reactsWith() description

Constructs a pattern where first and last small molecules are substrates to the same biochemical reaction.

Explicated Level 2

Pattern for a SmallMoleculeReference that has a member PhysicalEntity that participates in the same biological reaction than another small molecule linked to another SmallMoleculeReference.

```
CONSTRUCT {  
  ?smallMolecule1Ref abstraction:reactsWith ?smallMolecule2Ref .  
}  
WHERE {  
  ?reaction rdf:type/(rdfs:subClassOf*) bp3:BiochemicalReaction .  
  ?reaction bp3:left ?smallMolecule1 .  
  ?smallMolecule1 rdf:type bp3:SmallMolecule .  
  ?smallMolecule1 bp3:entityReference ?smallMolecule1Ref .  
  ?smallMolecule1Ref rdf:type bp3:SmallMoleculeReference .  
  
  ?reaction bp3:left ?smallMolecule2 .  
  ?smallMolecule2 rdf:type bp3:SmallMolecule .  
  ?smallMolecule2 bp3:entityReference ?smallMolecule2Ref .  
  ?smallMolecule2Ref rdf:type bp3:SmallMoleculeReference .  
  
  FILTER (STR(?smallMolecule1Ref) < STR(?smallMolecule2Ref))  
}
```


14 used-to-produce

Level 1: SIF description from PathwayCommons

A reaction consumes a small molecule to produce another small molecule.

Level 2: Paxtools pattern usedToProduce() description

Constructs a pattern where first small molecule is an input of a biochemical reaction that produces the second small molecule.

Explicated Level 2

Pattern for a SmallMoleculeReference that has a member PhysicalEntity that participates as a reactant in the same reaction than another small molecule that is produced. This product small molecule is linked to another SmallMoleculeReference.

```
CONSTRUCT {  
  ?smallMolecule1Ref abstraction:UsedToProduce ?smallMolecule2Ref .  
}  
WHERE {  
  ?reaction rdf:type/(rdfs:subClassOf*) bp3:Conversion .  
  ?reaction bp3:left ?smallMolecule1 .  
  ?reaction bp3:right ?smallMolecule2 .  
  ?smallMolecule1 rdf:type bp3:SmallMolecule .  
  ?smallMolecule1 bp3:entityReference ?smallMolecule1Ref .  
  ?smallMolecule1Ref rdf:type bp3:SmallMoleculeReference .  
  ?smallMolecule2 rdf:type bp3:SmallMolecule .  
  ?smallMolecule2 bp3:entityReference ?smallMolecule2Ref .  
  ?smallMolecule2Ref rdf:type bp3:SmallMoleculeReference .  
  
  FILTER (?smallMolecule1Ref != ?smallMolecule2Ref)  
}
```